

✓ app.py (With Full Comments)

```
# Import required modules

from flask import Flask, render_template, request, redirect

import mysql.connector


# Creating a Flask application object

app = Flask(__name__)


# Function to connect to MySQL Database

def get_connection():

    return mysql.connector.connect(

        host="localhost",

        user="root",      # Change if needed

        password="",      # Add your MySQL password if set

        database="college" # Your database name

    )


# ----- HOME PAGE -----

@app.route("/")

def home():

    # Display student registration form

    return render_template("register.html")


# ----- INSERT STUDENT DATA -----

@app.route("/register", methods=["POST"])

def register():

    # Receiving form data from register.html

    sid = request.form['sid']
```

```
sname = request.form['sname']
sbranch = request.form['sbranch']
smarks = request.form['smarks']
spho = request.form['spho']
```

```
# Connecting to the database
```

```
con = get_connection()
cur = con.cursor()
```

```
# Insert query
```

```
cur.execute("INSERT INTO student VALUES (%s, %s, %s, %s, %s)",
            (sid, sname, sbranch, smarks, spho))
con.commit() # Save changes permanently
con.close() # Close connection
```

```
# After registration go to view page
```

```
return redirect("/view")
```

```
# ----- DISPLAY STUDENTS -----
```

```
@app.route("/view")
```

```
def view():
```

```
    con = get_connection()
    cur = con.cursor()
```

```
# Fetch all student records
```

```
cur.execute("SELECT * FROM student")
data = cur.fetchall()
```

```
con.close()
```

```
# Passing data to HTML page
```

```
return render_template("view.html", students=data)
```

```
# ----- EDIT STUDENT PAGE -----
```

```
@app.route("/edit/<sid>")
```

```
def edit(sid):
```

```
    con = get_connection()
```

```
    cur = con.cursor()
```

```
# Fetch student by ID
```

```
cur.execute("SELECT * FROM student WHERE sid=%s", (sid,))
```

```
student = cur.fetchone()
```

```
con.close()
```

```
return render_template("edit.html", student=student)
```

```
# ----- UPDATE STUDENT DATA -----
```

```
@app.route("/update", methods=["POST"])
```

```
def update():
```

```
    sid = request.form['sid']
```

```
    sname = request.form['sname']
```

```
    sbranch = request.form['sbranch']
```

```
    smarks = request.form['smarks']
```

```
    spho = request.form['spho']
```

```
con = get_connection()
```

```
cur = con.cursor()

# Update query

cur.execute("""UPDATE student SET sname=%s, sbranch=%s, smarks=%s, spho=%s WHERE
sid=%s""",

            (sname, sbranch, smarks, spho, sid))

con.commit()

con.close()

return redirect("/view")

# ----- DELETE STUDENT DATA -----

@app.route("/delete/<sid>")
def delete(sid):
    con = get_connection()
    cur = con.cursor()

    # Delete query

    cur.execute("DELETE FROM student WHERE sid=%s", (sid,))

    con.commit()

    con.close()

    return redirect("/view")

# Run the Flask app

if __name__ == "__main__":
    app.run(debug=True)
```

✓ register.html (With Comments)

```
<!DOCTYPE html>

<html>

<head>

    <title>Register Student</title>

</head>

<body style="font-family:Arial; text-align:center;">

    <h2>Student Registration Form</h2>

    <!-- Form sends data to /register -->

    <form action="/register" method="POST">

        <input type="number" name="sid" placeholder="Student ID" required><br><br>
        <input type="text" name="sname" placeholder="Student Name" required><br><br>
        <input type="text" name="sbranch" placeholder="Branch" required><br><br>
        <input type="number" name="smarks" placeholder="Marks" required><br><br>
        <input type="text" name="spho" placeholder="Phone Number" required><br><br>

        <button type="submit">Register</button>

    </form>

    <br>

    <!-- Link to view all data -->

    <a href="/view">View Student List</a>

</body>

</html>
```

✓ view.html (With Comments)

```
<!DOCTYPE html>

<html>

<head>

  <title>View Students</title>

</head>

<body style="font-family:Arial; text-align:center;">


<h2>Registered Students List</h2>


<!-- Table to display student records -->
<table border="1" align="center" cellpadding="10">

  <tr>

    <th>ID</th>

    <th>Name</th>

    <th>Branch</th>

    <th>Marks</th>

    <th>Phone</th>

    <th>Actions</th>

  </tr>


  <!-- Loop through students and display rows -->
  {% for s in students %}

  <tr>

    <td>{{ s[0] }}</td>

    <td>{{ s[1] }}</td>

    <td>{{ s[2] }}</td>

    <td>{{ s[3] }}</td>
```

```

<td>{{ s[4] }}</td>

<td>

  <a href="/edit/{{ s[0] }}">Edit</a> |

  <a href="/delete/{{ s[0] }}" onclick="return confirm('Do you really want to
delete?');">Delete</a>

</td>

</tr>

{% endfor %}

</table>

<br>

<a href="/">Register Another Student</a>

</body>

</html>

```

✓ edit.html (With Comments)

```

<!DOCTYPE html>

<html>

<head>

  <title>Edit Student</title>

</head>

<body style="font-family:Arial; text-align:center;">

<h2>Edit Student Information</h2>

<!-- Sending updated data to /update -->

```

```
<form action="/update" method="POST">
```

```
<!-- sid cannot be changed -->
```

```
<input type="text" name="sid" value="{{ student[0] }}" readonly><br><br>
```

```
<input type="text" name="sname" value="{{ student[1] }}" required><br><br>
```

```
<input type="text" name="sbranch" value="{{ student[2] }}" required><br><br>
```

```
<input type="number" name="smarks" value="{{ student[3] }}" required><br><br>
```

```
<input type="text" name="spfo" value="{{ student[4] }}" required><br><br>
```

```
<button type="submit">Update</button>
```

```
</form>
```

```
<br>
```

```
<a href="/view">Back to Student List</a>
```

```
</body>
```

```
</html>
```